1. 12 members were present at a board meeting. Each member shook hands will all of the other memebers before and after the meeting. How many hand shakes were there? **(P & C) NIMCET-2008**

(a)118	(b)127
(c)132	(d)264

2. In an objective type examination, 120 objective type questions are there : each with 4 options P, Q, R and S. A candidate can choose either one of these options or can leave the question unanswered. How many different ways exist for answering this question paper? **(P & C) NIMCET-2008**

(a)5^{120}	(b)4^{120}
(c)120^{5}	(d)120^{4}

3. Steel Express runs between Tata Nagar and Howrah and has five stoppages in between Find the number of different kinds of one-way second class ticket that Indian Railway will have to print to service all types of passenger who might travel by Steel Express? **(P & C) NIMCET-2010**

(a)49	(b) 42
(c) 21	(d) 7

4. In how many different ways can the letters of the word "DETAIL" be arranged in such a way that the vowels occupy only the odd positions? **(P & C) NIMCET-2011**
(a)32 (b)36 (c)48 (d)60

5. From a group of 7 men and 6 women, a committee of 5 persons with more males than females is to be formed. In how many ways can this be done? **(P & C) NIMCET-2017**
(a)564 (b)645 (c)735 (d)756

ANSWER

1	**c**	**2**	**a**	**3**	**b**	**4**	**b**	**5**	**d**

SOLUTION:

1. **(c)**
Total number of hand shakes will be $2({}^{12}C_2) = 132$

2. **(a)**
Each question has 5 options, as it can be answered as A B, C, D or left also.
Now, all 120 questions will have 5 independent options.
Hence, total number of required ways
$= 5 \times 5 \times 5 \ldots \times 5$ (120 times) $= 5^{120}$.

3. **(b)**Therefore the total number of different one way tickets:
$= {}^{7}C_2 \times 2 = 42$

4. **(b)**

E		A		I	
1	2	3	4	5	6

Hence, number of ways = 3! x 3! =3x2x3x2=36

5. **(d)** There are three number of ways by which we can form the committee.
(i) 3 Males and 2 Females
(ii) 4 Males and 1 Female
(iii) 5 Males +0 Female
Hence, total number of ways
$= {}^{7}C_3\,{}^{6}C_2 + {}^{7}C_4\,{}^{6}C_1 + {}^{7}C_5\,{}^{6}C_0$
$= \frac{7\times6\times5}{3\times2\times1} \times \frac{6\times5}{2\times1} + \frac{7\times6\times5}{3\times2\times1} \times \frac{6}{1} + \frac{7\times6}{2\times1} \times 1$
$= 525 + 210 + 21 = 756$
Hence, this can be done by 756 ways.